Week 4 — Market Sizing Methodology Memo

inGen Dynamics service-robotics portfolio · TAM / SAM / SOM across five verticals (Fari, Senpai, Sentinel Prime AI, Aido Rover, Aido Humanoid)

1. Definitions

TAM (Total Addressable Market) — global revenue if every relevant buyer used a product in the category. Taken from published market estimates. **SAM (Serviceable Available Market)** — the slice InGen could realistically serve, here approximated as the US portion of TAM (InGen is US-based; US share per regional splits in the sources). **SOM (Serviceable Obtainable Market)** — what InGen could capture in ~3–5 years given competition and go-to-market reality.

2. Two independent methods (we run both)

Top-down: start from a published global market size, take a LOW and a HIGH anchor from different reputable firms (because estimates disagree), multiply by US share for SAM, then apply an illustrative 2–8% early-capture band for SOM. **Bottom-up:** SOM = addressable units × serviceable % × penetration % × ASP, with low/base/high on every driver. The two methods are then reconciled per vertical; the goal is a defensible *range*, not a single number.

3. Why everything is a range (honesty note)

Published estimates for these categories disagree substantially. Examples found in sourcing: humanoid robots range from about $290M to $4.89B for 2024–2025 depending on scope and firm; security robots span roughly $4.7B to $19B (largely because some totals bundle defense/UAV); narrow 'security patrol robot' is cited as low as $153M. We therefore never report a single spuriously-precise figure: each vertical carries low/high anchors, bottom-up low/base/high, named sources, and a sensitivity (tornado) chart. **Humanoid (Aido Humanoid)** and **outdoor patrol (Aido Rover)** are the least certain — one pre-commercial, one definitionally narrow — and are presented as scenario ranges rather than forecasts.

4. Per-vertical results (summary)

Vertical / product	TAM (global)	SAM (US)	Bottom-up SOM (low–base–high)
Fari (Eldercare)	$2.70B–$3.40B	$810M–$1.02B	$7M / $104M / $775M
Senpai (Education)	$1.40B–$2.20B	$350M–$550M	$0M / $4M / $34M
Sentinel Prime AI (Indoor Security)	$4.69B–$19.07B	$1.78B–$7.25B	$6M / $108M / $630M
Aido Rover (Outdoor Patrol)	$153M–$2.00B	$61M–$800M	$1M / $32M / $252M
Aido Humanoid (Humanoid)	$290M–$4.89B	$139M–$2.35B	$0M / $4M / $72M

Full build (every driver, both methods, reconciliation, sources) is in **market_sizing_workbook.xlsx**; assumptions in **data/week04/market_sizing_assumptions.csv**; sensitivity charts are the five **tornado_*.png** files.

5. Sensitivity (which assumption matters most)

For each vertical a one-at-a-time tornado analysis varies each driver to its low/high while holding the others at base. The widest bar is the dominant uncertainty: **penetration %** dominates eldercare; **ASP** dominates indoor security (MaaS-vs-guard economics) and is co-dominant for humanoid; serviceable-site count and penetration drive the narrow outdoor-patrol niche. This tells InGen where a single better data point would most tighten the estimate.

6. Data lineage

Demand-side bases (US 65+ population, K-12 enrollment, security/warehouse labor, robotics-research momentum) come from the Week 3 DuckDB warehouse (Census, BLS, NCES, World Bank, OpenAlex) — versioned and sha256-logged. Market totals come from public market-research summaries (Grand View, Mordor, Precedence, Fortune Business Insights, MarketsandMarkets, MRFR, others), retrieved 2026-06-01. Every per-vertical tab lists its sources.

Sources: see each workbook tab and data/week04/market_sizing_assumptions.csv. Market totals are public market-research summaries (vendor estimates vary; treat as ranges). Demand bases from the Week 3 public-data warehouse. Prepared by Ziheng Wang · inGen Data Analyst internship · Week 4 · 2026-06-01. Figures are planning estimates, not investment advice; verify vendor reports before external use.